

Math Made Clear

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About the Author / 저자 소개

Jueon Kim and Yeoeun Kim are students who have lived in Korea, China, and the United States, where they experienced firsthand how language and culture shape learning. Through their nonprofit organization, GYCO, they learned the value of sharing knowledge by serving vulnerable communities. They bring music to seniors, the homeless, and cancer patients through monthly performances, and they send books to refugees and students in Africa to support education. Their commitment to service has inspired many peers, and this math book reflects their own journey as Korean immigrants adapting to new languages and education systems. They hope it helps students overcome language barriers, see math as approachable, and use it as a tool for confidence and opportunity.

저자 소개

김주언, 김여은 두 학생은 한국, 중국, 미국에서 생활하며 언어와 문화가 학습에 어떤 영향을 주는지 직접 경험한 학생입니다. 새로운 환경에 적응해야 했을 뿐 아니라 새로운 문화도 배워야 했고 그 과정 속에서 이 학생들은 잘 배워서 나누는 사람이 되는 것을 비영리 단체 GYCO를 통하여 중요한 가치로 배우게 되었습니다. 음악연주를 직접 보러 올 수 없으신 사회의 연약한 계층 (시니어, 노숙자, 암환자) 에게 한달에 한 번씩 연주로 찾아가고, 교육이 필요한 난민과 아프리카의 학생들에게 책을 보내어 교육받을 수 있게 도왔습니다. 많은 시니어들에게 찾아가 연주가 끝나고 함께 이야기를 나누며 그들에게 위로가 되어드렸고, 노숙자분들께 필요한 물품도 준비해서 나눠드리는 활동도 지속적으로 해오고 있습니다. 함께 봉사하는 학생들에게도 이들의 성실함과 열정과 나눔은 늘 도전이 되었습니다. 이 수학책은 한국 이민자로서 새로운 언어와 교육 제도에 적응하면서 느낀 두 학생의 어려움을 바탕으로, 같은 길을 걷는 학생들이 수학을 더 쉽고 친근하게 느끼도록 돕기 위해 시작되었습니다. 언어의 장벽으로 힘들어 하는 학생들이 수학을 어려움으로 느끼지 않고, 변화된 상황에서 뛰어난 수학 실력으로 자신감을 갖고 기회를 넓혀주는 도구가 되기를 희망하며 책을 만든 귀한 학생들입니다.

Acknowledgements / 감사의 글

I am deeply grateful to my loving parents, teachers, and friends who encouraged and supported me throughout the process of creating this book. This project began from my own experience of struggling with different terms and units when I first came to the United States, and I sincerely hope it will help Korean students facing similar challenges. A special thanks goes to GYCO, which has always taught me that learning and sharing are the greatest values, and has given me the opportunity to put those values into practice.

감사의 말

이 책을 만드는 과정에서 격려와 지지를 보내주신 사랑하는 부모님, 선생님, 친구들에게 깊은 감사를 드립니다. 처음 미국에 왔을 때 용어와 단위가 한국과 달라 어려웠던 저의 경험에서 시작된 이 책이 같은 어려움을 겪고 있는 한국 친구들에게 도움이 되길 진심으로 바랍니다.. 늘 배움과 나눔을 가장 큰 가치로 가르쳐 주시고 실천하게 해주신 GYCO에 특별한 감사를 전합니다

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Endorsements / 추천의 글

이 책은 우리 한글학교 학생들의 배움을 향한 사랑과 정성이 모여 탄생한 소중한 열매입니다. 특히, 자원봉사 학생이 시간을 내어 영어 수학 용어를 한국어로 정리한 이 책은 또래 학생들을 돕고자 하는 따뜻한 마음에서 시작되었습니다. 이 책을 통해 학생들이 수학을 조금 더 친근하게 느끼고 나아가 한국어로 학문을 접하는 기쁨을 누리기를 바랍니다. 작은 노력이 큰 빛이 되어 다른 학생들의 길을 밝혀주듯 이 책이 많은 학생들에게 든든한 길잡이가 되기를 기대합니다. 정성과 사랑으로 이 귀한 작업을 완성한 김주언, 김여은 학생에게 깊은 감사와 축복의 마음을 전합니다.

— 잔스크릭한국학교 교장 황영희.

자신의 어려웠던 경험을 생각하며 같은 어려움을 겪고 있는 학생들을 돕기 위해 수학용어 책을 만들 생각을 하고, 생각에 그친 것이 아니라 바쁜 시간속에서 책을 쓰기 위해 자료를 수집하고 노력해 완성해 낸 너무 대견하고 자랑스러운 학생들입니다. “배움과 나눔”이라는 가치를 배워서 이타적인 삶을 어린나이에 실천한 김주언, 김여은 학생이 너무 자랑스롭습니다. 앞으로 본인들이 전공할 학문에서 더 깊이 있게 공부하고 더 많은 사람들에게 지식과 마음을 나눌 수 있는 귀한 학생들이 될 것을 믿고 그 찬란하게 빛날 앞길에 축복의 마음을 더합니다.

— GYCO 회장 김현정

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Introduction / 서문

Welcome to this math book designed especially for Korean-American students learning Korean! This book is organized to help you build a strong foundation in mathematics while improving your English skills. Each section covers a key topic, starting with clear explanations and followed by examples, practice questions, and helpful tips.

The book is divided into major sections such as Numbers & Operations, Expressions & Equations, Data & Statistics, and more. Within each section, you will find step-by-step guides, vocabulary lists, and real-world applications. Special features include bilingual glossaries, common mistakes to avoid, and mini quizzes to check your understanding.

To get the most out of this book, we recommend reading each section carefully, trying the examples on your own, and using the vocabulary lists to strengthen your math English. Do not hesitate to revisit earlier sections if you need a refresher. Remember, learning math in a new language takes time and practice—be patient with yourself and celebrate your progress!

We hope this book becomes a valuable companion on your journey to mastering math and Korean together.

한국어를 배우는 미국 학생들을 위해 특별히 설계된 이 수학책에 오신 것을 환영합니다! 이 책은 여러분의 한국어 실력을 향상시키는 동시에 수학에 대한 탄탄한 기초를 쌓을 수 있도록 구성되어 있습니다. 각 섹션으로 나누어 핵심 주제를 다루며, 알기 쉽고 정확한 설명으로 시작하여 예시, 연습 문제, 그리고 실질적으로 유용한 팁으로 이어집니다.

이 책은 수와 연산, 식과 방정식, 데이터와 통계 등의 주요 섹션으로 나뉩니다. 각 섹션내에서는 단계별 가이드, 미국 수학 어휘 목록, 실제 응용 사례를 찾을 수 있습니다. 특별히 한국에서 수학을 했던 학생이 미국수학을 접할때 알아야 하는 이중언어 용어집, 피해야 할 일반적인 실수, 그리고 이해도를 확인할 수 있는 미니 퀴즈가 있습니다.

이 책을 최대한 활용하려면, 각 섹션을 주의 깊게 읽고, 예시를 직접 풀이 해보며, 어휘 목록을 사용하여 수학에서 사용하는 영어를 잘 익혀야 합니다. 복습이 필요하면 이전 섹션을 다시 찾아보는 것을 주저하지 마세요. 기억하세요, 새로운 언어로 수학을 배우는 것은 시간과 연습이 필요합니다 —자기 스스로에게 인내심을 가지고 꾸준히 연습하면 어렵지 않습니다!

우리는 이 책이 여러분의 수학과 한국어를 함께 마스터하는 여정에 소중한 동반자가 되기를 희망합니다.

Chapter 1

Numbers & Operations / 수와 연산

Welcome! In this book, you'll explore the world of numbers and how we use them every day. You'll learn about place value, decimals, fractions, and more. Each Section includes explanations, examples, and practice problems to help you master these important math skills.

1.1 Place Value / 자리 값

In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number. This idea is called place value. Each place in a number is ten times the value of the one to its right. A number like 4275 is not just four digits. It is a structure: $4000 + 200 + 70 + 5$. We do not read numbers one digit at a time. We read them as a system, where each part works because of its position.

Place value turns a few digits into large or small numbers. Take 9836. The 8 is not just an 8. It is 800, because it sits in hundreds of places. In 5104, the 0 adds nothing but still matters. It shows that there are no tens. In 12345, the 2 sits in the thousands place and means 2000. These examples may seem simple, but they show a key truth: digits mean more when you know where they stand.

“In 5104, the 0 adds nothing but still matters.”

5104에서 0은 값을 더하지 않지만 중요한 역할을 한다. 십의 자리가 비어 있음을 보여주며 숫자의 위치를 유지하는 자리 표시자이다.

Place value is more than vocabulary. It affects how we add, subtract, estimate, and think about numbers. When we know what each digit means, we can break numbers apart and put them back together. We stop guessing and start understanding. That shift helps us see math not as a list of rules but as a system that makes sense.

“Each place in a number is ten times the value of the one to its right.”

각 자리값은 바로 오른쪽 자리의 10배이다. 예: 십의 자리는 일의 자리보다 10배 크다.

1.1.1 What is Place Value? / 자리 값이란?

Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.

To truly grasp the concept of place value, for example, let's take a closer look at the number 4275. Here, the digit 4 is found in the thousands place, which means it actually represents 4000, not just 4. Moving to the right, the 2 is in the hundreds place, so it stands for 200. The 7, sitting in the tens place, is worth 70, and finally, the 5 in the ones place simply means 5. This system, where each digit's value increases by a factor of ten as you move left, is what makes our decimal system so powerful and flexible.

To help visualize this, consider the following place value chart:

Place	Value
Thousands	1000
Hundreds	100
Tens	10
Ones	1

Let's see how this works in practice. If we write 4275 in expanded form, we get:

$$4275 = 4000 + 200 + 70 + 5$$

Now, imagine you are looking at the number 9836. You might wonder, what is the value of the digit 8 in this number? Since the 8 is in the hundreds place, its value is 800. This simple observation shows how the position of a digit determines its true value. Similarly, if you break down 5104 into its expanded form, you see it as $5000 + 100 + 0 + 4$, making it clear what each digit stands for. In another example, the digit 2 in 12345 is in the thousands place, so it represents 2000. Understanding these positions helps us see the true value of every digit in a number, which is a fundamental skill in mathematics.

1.2 Decimals and Fractions / 소수와 분수

Sometimes, a whole number is too much. We want to describe something in parts: half a sandwich, a third of a mile, seventy-five hundredths of a dollar. That's when we use fractions and decimals. There are two ways of saying the same thing: less than one. Fractions use numerators and denominators. Decimals use place value. But both express parts of a whole.

Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$. The fraction $\frac{3}{10}$ becomes 0.3. The decimal 0.6 becomes $\frac{6}{10}$, which is $\frac{3}{5}$. Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.

“Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$.”

소수와 분수는 서로 다른 언어가 아니라 서로 변환할 수 있는 표현 방식이다. 예: $0.75 = \frac{3}{4}$.

1.2.1 What are Decimals and Fractions? / 소수와 분수란?

As we move beyond whole numbers, we encounter quantities that are less than one. These are often represented as fractions or decimals. A **fraction** shows a part of a whole, such as $\frac{1}{2}$ or $\frac{3}{4}$. On the other hand, a **decimal** expresses the same idea using a decimal point, like 0.5 or 0.75. Both forms are just different ways to describe the same concept.

For example, consider the decimal 0.75. This can be written as the fraction $\frac{75}{100}$, which simplifies to $\frac{3}{4}$:

$$0.75 = \frac{75}{100} = \frac{3}{4}$$

Let's try converting between these forms. If you have the fraction $\frac{3}{10}$, you can write it as the decimal 0.3. Conversely, if you start with the decimal 0.6, you can express it as a fraction: $0.6 = \frac{6}{10} = \frac{3}{5}$ in simplest form. Being able to move between decimals and fractions is a key skill that will help you in many areas of math.

Fractions / 분수

분수에는 진분수, 가분수, 대분수가 있다. 진분수는 $\frac{1}{4}, \frac{2}{4}, \frac{3}{4}$ 와 같이 분자가 분모보다 작은 분수다. 가분수는 $\frac{4}{4}, \frac{5}{4}$ 와 같이 분자가 분모와 같거나 분모보다 큰 분수이고 대분수는 $1\frac{1}{4}, 2\frac{3}{4}$ 과 같이 자연수와 진분수로 이루어진 분수다. 대분수 $1\frac{1}{4}$ 을 가분수로 나타내면 $\frac{5}{4}$ 이다. 가분수 $\frac{9}{4}$ 를 대분수로 나타내면 $2\frac{1}{4}$ 이다.

Fractions include proper fractions, improper fractions, and mixed numbers. A proper fraction has a numerator smaller than the denominator, like $\frac{1}{4}, \frac{2}{4}, \frac{3}{4}$. An improper fraction has a numerator equal to or greater than the denominator, like $\frac{4}{4}, \frac{5}{4}$. A mixed number is a whole number with a proper fraction, like $1\frac{1}{4}, 2\frac{3}{4}$. The mixed number $1\frac{1}{4}$ written as an improper fraction is $\frac{5}{4}$. The improper fraction $\frac{9}{4}$ written as a mixed number is $2\frac{1}{4}$.

Addition of Proper Fractions / 진분수의 덧셈

분모가 같은 진분수끼리의 덧셈은 분모는 그대로 쓰고, 분자끼리 더하면 된다. 이때 덧셈의 결과가 가분수이면 대분수로 나타낸다.

When adding proper fractions with the same denominator, keep the denominator the same and add the numerators. If the result is an improper fraction, express it as a mixed number.

$$\frac{2}{4} + \frac{3}{4} = \frac{2+3}{4} = \frac{5}{4} = 1\frac{1}{4}$$

Addition of Mixed Numbers / 대분수의 덧셈

분모가 같은 대분수의 덧셈을 할 때는 자연수는 자연수끼리, 분수는 분수끼리 더한다. 이때 분수끼리의 합이 가분수이면 대분수로 고쳐서 1을 받아올려야 한다. 또 다른 방법으로는 대분수를 가분수로 나타내어 계산한 후 다시 대분수로 나타낸다.

When adding mixed numbers with the same denominator, add whole numbers with whole numbers, and fractions with fractions. If the sum of the fractions is an improper fraction, convert it into a mixed number and carry over 1. Another method is to convert the mixed number into an improper fraction, calculate, and then convert back into a mixed number.

$$1\frac{4}{5} + 1\frac{3}{5} = (1 + 1) + \frac{4+3}{5} = 2 + \frac{7}{5} = 2 + 1\frac{2}{5} = 3\frac{2}{5}$$

Subtraction of Fractions / 분수의 뺄셈

분모는 그대로 두고, 분자끼리 빼면 된다.

Keep the denominator the same, and subtract the numerators.

$$\frac{8}{21} - \frac{5}{21} = \frac{8-5}{21} = \frac{3}{21}$$

자연수에서 1을 진분수의 분모와 같은 분수로 고친다. 자연수의 분모는 그대로 쓰고, 분자끼리 빼면 된다.

Convert 1 from the whole number into a fraction with the same denominator as the proper fraction.

$$3 - \frac{1}{4} = 2 + \frac{4}{4} - \frac{1}{4} = 2 + \frac{4-1}{4} = 2 + \frac{3}{4} = 2\frac{3}{4}$$

Addition of Decimals / 소수의 덧셈

자릿수가 같은 소수의 덧셈은 소수점 찍는 자리에만 주의한다면 자연수의 덧셈과 같다.

Adding decimals with the same number of decimal places is just like adding whole numbers, as long as you pay attention to the decimal point.

$$0.3 + 0.5 = 0.8(3 + 5 = 8), \quad 0.24 + 0.51 = 0.75(24 + 51 = 75), \quad 23.6 + 19.4 = 43(236 + 194 = 430)$$

소수점끼리 자리를 맞춘다. 자연수의 덧셈과 같은 방법으로 받아올림하여 계산한다. 합에 소수점을 맞춰 찍는다. 덧셈을 하고 나서 소수점 끝자리의 0은 생략할 수 있다.

Line up the decimal points. Add and carry just like with whole numbers. Place the decimal point in the sum. Drop trailing zeros.

Subtraction of Decimals / 소수의 뺄셈

자릿수가 같은 소수의 뺄셈은 소수점 찍는 것만 주의한다면 자연수의 뺄셈과 같다.

Subtracting decimals with the same number of decimal places is just like subtracting whole numbers, as long as you pay attention to the decimal point.

$$0.7 - 0.2 = 0.5(7 - 2 = 5), \quad 0.48 - 0.15 = 0.33(48 - 15 = 33), \quad 31.8 - 14.8 = 17(318 - 148 = 17)$$

자릿수가 다른 소수의 뺄셈을 할 때는 자연수의 뺄셈과 같이 계산하되 특히 자릿수를 맞추는 것에 주의해야 한다.

When subtracting decimals with different numbers of decimal places, calculate as with whole numbers, but be careful to line up the decimal places.

$$4.6 - 2.18 = 2.42(460 - 218 = 242), \quad 6.45 - 1.3 = 5.15(645 - 130 = 515), \quad 54.13 - 3.6 = 50.53(5413 - 360 = 5053)$$

Multiplication of Fractions / 분수의 곱셈

단위분수는 $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ 같이 분자가 1인 분수이고, 진분수의 한 종류이다. 단위분수 $\times$ 단위분수는 분자는 그대로 두고, 분모끼리 곱하여 계산한다. 진분수 $\times$ 진분수는 분자는 분자끼리, 분모는 분모끼리 곱하여 계산한다. 이때 약분이 되면 약분을 해서 기약분수로 나타낸다.

Unit fractions like $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ have numerator 1. For unit $\times$ unit: keep numerator 1 and multiply denominators. For proper $\times$ proper: multiply numerator $\times$ numerator and denominator $\times$ denominator. Reduce if possible.

$$\frac{1}{4} \times \frac{1}{8} = \frac{1}{4 \times 8} = \frac{1}{32}$$

$$\frac{3}{4} \times \frac{1}{6} = \frac{3 \times 1}{4 \times 6} = \frac{3}{24} = \frac{1}{8}$$

With Mixed Numbers / 대분수와 곱셈 (진분수 $\times$ 대분수), (가분수 $\times$ 대분수): 대분수를 모두 가분수로 고쳐서 계산한다. 약분이 되면 약분을 하고, 계산 결과가 가분수이면 대분수로 고친다.

Convert mixed numbers to improper fractions, multiply, simplify, then convert back if needed.

$$\frac{2}{3} \times 1\frac{2}{4} = \frac{2}{3} \times \frac{6}{4} = \frac{12}{12} = 1, \quad \frac{9}{7} \times 1\frac{3}{4} = \frac{9}{7} \times \frac{7}{4} = \frac{63}{28} = 2\frac{1}{4}$$

1.3 Comparing and Ordering Numbers / 수의 비교와 정렬

Math is not just about knowing what numbers are. It is about knowing how they relate. We compare. We rank. We ask, which is bigger? Which is smaller? Which is equal? Symbols like $>$, $<$, and $=$ help us answer. But first, we have to understand the values.

Take 5.03 and 5.3. They may look close. But 5.3 is larger. Now try comparing fractions: $\frac{2}{5}$ and $\frac{3}{7}$. Convert them to decimals. $\frac{2}{5}$ is 0.4. $\frac{3}{7}$ is about 0.428. So $\frac{3}{7}$ is greater. This kind of comparison matters in real life—from prices to measurements to data.

Ordering numbers builds the same skill. Imagine you are given 0.2, $\frac{1}{3}$, 0.25, and 0.199. Which is the greatest? Which is least? Convert, compare, and you find the order. This process teaches more than math. It teaches precision, logic, and clarity.

“Math is not just about knowing what numbers are. It is about knowing how they relate.”

수학은 단순히 수를 아는 것에서 끝나지 않고, 수 사이의 관계를 이해하는 것이다.

“We compare. We rank. We ask, which is bigger? Which is smaller?”

수를 비교한다. 순서를 정한다. 뭐가 크고, 뭐가 작은지 묻는다.

“Symbols like $>$, $<$, and $=$ help us answer.”

$>$, $<$, $=$ 기호는 수의 크기를 비교하는 데 사용된다.

1.3.1 How do we compare and order numbers? / 어떻게 비교하고 정렬하나요?

In mathematics, it's important not only to understand numbers, but also to compare them and put them in order. We use symbols like $>$ (greater than), $<$ (less than), and $=$ (equal to) to show how numbers relate to each other. For example, if you are given two numbers, you can decide which is larger or smaller, or if they are the same.

Let's look at a few examples. Suppose you want to compare 5.03 and 5.3. Since $5.03 < 5.3$, we use the less than symbol. Now, what if you have two fractions, $\frac{2}{5}$ and $\frac{3}{7}$? By converting them to decimals, you find that $\frac{2}{5} = 0.4$ and $\frac{3}{7} \approx 0.428$, so $0.4 < 0.428$, which means $\frac{2}{5} < \frac{3}{7}$.

Ordering numbers is also a useful skill. Imagine you are asked to arrange the numbers 0.2, $\frac{1}{3}$, 0.25, 0.199 from greatest to least. By converting them all to decimals, you see that $\frac{1}{3}$ (approximately 0.333) is the largest, followed by 0.25, then 0.2, and finally 0.199. Sometimes, you may need to compare a fraction and a decimal, such as $\frac{4}{9}$ and 0.44. Since $\frac{4}{9} \approx 0.444$, it is slightly greater than 0.44. These skills are essential for working with numbers in real-life situations.

1.4 Factors and Multiples / 약수와 배수

Some numbers fit together cleanly. Others do not. Factors and multiples explain why. A factor is a number that divides evenly. A multiple is what you get when you multiply. These two ideas help us break numbers apart and build them back up.

If you list the factors of 12, you get 1, 2, 3, 4, 6, and 12. Each divides 12 without a remainder. The multiples of 5 are 5, 10, 15, 20, and so on. These ideas may seem mechanical, but they are powerful. They lay the groundwork for division, fractions, and algebra.

“A factor is a number that divides evenly. A multiple is what you get when you multiply.”

factor = 약수. 약수는 어떤 수를 나누었을 때 나머지가 없이 나누어 떨어지게 하는 수이다. 예: 3은 6의 약수이다 $6/3=2$
multiple = 배수. 배수는 어떤 정수에 다른 정수를 곱하여 얻은 수이다. 예: 6은 3의 배수이다 $3*2=6$

1.4.1 What are Factors and Multiples? / 약수와 배수란?

As you continue your journey in mathematics, you'll often need to break numbers down or build them up. This is where factors and multiples come in. A **factor** is a number that divides another number exactly, leaving no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12, because each of these divides 12 evenly. On the other hand, a **multiple** is what you get when you multiply a number by an integer. For instance, the multiples of 5 are 5, 10, 15, 20, and so on.

Let's see how this works with a few examples. If you want to find all the factors of 24, you list out 1, 2, 3, 4, 6, 8, 12, and 24. If you're asked for the first 6 multiples of 8, you simply multiply: 8, 16, 24, 32, 40, and 48. Sometimes, you might wonder if one number is a factor of another. For example, is 9 a factor of 81? Since $81 \div 9 = 9$ with no remainder, the answer is yes. Understanding factors and multiples is crucial for topics like division, fractions, and finding patterns in numbers.

Factors and Multiples / 약수와 배수

Factors / 약수 곱셈식 ★을 어떤 수로 나누었을 때 나누어떨어지게 하는 어떤 수를 ★의 약수라고 한다. ★의 약수에는 1과 자기 자신이 항상 있다.

A factor of ★ divides ★ evenly. 1 and ★ are always factors of ★.

6의 약수 $\rightarrow$ 1, 2, 3, 6

Multiples / 배수 ★의 배수는 1배, 2배, 3배, ... 한 수를 말한다. 곱셈식에서 $\Pi \times n$ 은 ★의 배수이다. ★의 배수에는 자기 자신이 항상 있다.

A multiple of ★ is found by multiplying ★ by 1, 2, 3, and so on. Every number is also its own multiple.

5의 배수 $\rightarrow$ 5, 10, 15, 20, 25, ...

GCF and LCM / 최대공약수와 최소공배수

By Listing / 목록으로 구하기 두 수 각각의 약수를 구한 후 공통인 약수가 공약수, 공약수 중에서 가장 큰 수가 최대공약수이다.

List factors of each; shared factors are common factors; the largest is the GCF.

12의 약수 $\rightarrow$ 1, 2, 3, 4, 6, 12

18의 약수 $\rightarrow$ 1, 2, 3, 6, 9, 18

공약수 = 1, 2, 3, 6 $\Rightarrow$ 최대공약수 = 6

LCM by Multiples / 배수로 구하기 두 수의 배수를 각각 구한다. 공통인 배수는 공배수, 그중에서 가장 작은 수가 최소공배수이다.

$$4\text{의 배수} \rightarrow 4, 8, 12, 16, 20, \dots$$

$$6\text{의 배수} \rightarrow 6, 12, 18, 24, \dots$$

$$\text{공배수} = 12, 24, 36, \dots \Rightarrow \text{최소공배수} = 12$$

Prime Factorization / 소인수분해로 구하기 두 수를 1이 아닌 가장 작은 수들의 곱으로 나타낸다. 공통으로 들어 있는 수들의 곱이 최대공약수, 공통으로 들어있지 않은 수들까지 곱한 것이 최소공배수이다.

Express each as product of primes. Product of shared primes = GCF; product of all primes (max powers) = LCM.

$$18 = 2 \times 3 \times 3, \quad 27 = 3 \times 3 \times 3$$

$$\text{GCF} = 3 \times 3 = 9, \quad \text{LCM} = 2 \times 3 \times 3 \times 3 = 54$$

Making Like Fractions / 약분과 통분

Simplifying / 약분 같은 분수로 나타내기: 분모와 분자를 그들의 공약수로 나누는 것을 약분이라고 한다.

Simplify by dividing numerator and denominator by a common factor.

$$\frac{9}{81} = \frac{1}{9}$$

Common Denominators / 통분 두 분모의 곱을 이용하거나, 최소공배수를 이용해서 분모를 같게 만드는 것을 통분이라고 한다.

Make denominators equal by multiplying or using the LCM.

$$\left(\frac{4}{9}, \frac{5}{6}\right) \rightarrow \left(\frac{4 \cdot 2}{9 \cdot 2}, \frac{5 \cdot 3}{6 \cdot 3}\right) = \left(\frac{8}{18}, \frac{15}{18}\right)$$

1.5 Prime and Composite Numbers / 소수와 합성수

Some numbers stand alone. Others are made of parts. Prime numbers have only two factors: 1 and themselves. Composite numbers have more. This is not just a label—it is a clue to how numbers work.

2, 3, 5, 7, 11, 13, and 17 are prime. They cannot be broken down. But 15 is composite: it splits into 3 and 5. Learning to recognize prime and composite numbers helps us factor numbers, simplify fractions, and understand the building blocks of math.

“Prime numbers have only two factors: 1 and themselves. Composite numbers have more.”

prime number = 소수. 소수는 1과 자기 자신 두 개의 약수만 가지는 수이다. 예: 2, 3, 5, 7.

composite number = 합성수. 합성수는 두 개보다 많은 약수를 가지는 수이다. 예: 10, 15, 20.

1.5.1 What are Prime and Composite Numbers? / 소수와 합성수란?

As you explore numbers further, you’ll discover that some numbers have only two factors, while others have many. A **prime number** is a number greater than 1 that has exactly two distinct factors: 1 and itself. For example, 2, 3, 5, 7, 11, 13, 17, and 19 are all prime numbers under 20. In contrast, a **composite number** has more than two factors. For instance, 15 is composite because it can be divided by 1, 3, 5, and 15.

Let’s practice identifying these. Consider the numbers 14, 17, 21, 23, and 25. Among these, 17 and 23 are prime because they have only two factors. The others—14, 21, and 25—are composite. If you look at all the composite numbers between 10 and 20, you’ll find 10, 12, 14, 15, 16, 18, and 20. It’s also important to note that the number 1 is neither prime nor composite, since it has only one factor: itself. Recognizing prime and composite numbers is a foundational skill for understanding more advanced math concepts like prime factorization and divisibility.

1.6 Greatest Common Factor & Least Common Multiple / 최대공약수와 최소공배수

Sometimes two numbers meet in the middle. The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple. These ideas sound opposite, but they both help us solve problems with shared values.

Take 12 and 18. Their GCF is 6 since it is the biggest number that divides them both. Their LCM is 36 which is the smallest number they both fit into. We find these using prime factorization. Once you learn how, you see how numbers connect. And once you see the connection, math becomes clearer.

“The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple.”

GCF = 최대공약수. 1보다 큰 자연수 중에서 1과 자기 자신만을 약수로 갖는 소수가 아닌 수. 예시: 12와 18의 최대공약수는 6

LCM = 최소공배수. 두 개 이상의 여러 수에 공통으로 존재하는 배수들 중에서 가장 작은 수 예시: 12와 18의 최소공배수는 36

1.6.1 What are GCF and LCM? / 최대공약수와 최소공배수란?

When working with two or more numbers, it's often useful to find numbers they have in common. The **Greatest Common Factor (GCF)** is the largest number that divides two numbers exactly, while the **Least Common Multiple (LCM)** is the smallest number that both numbers divide into evenly.

To find the GCF and LCM, you can use prime factorization. For example, let's find the GCF and LCM of 12 and 18. The prime factorization of 12 is $2^2 \times 3$, and for 18 it's 2×3^2 . The GCF is the product of the lowest powers of common primes: $2^1 \times 3^1 = 6$. The LCM is the product of the highest powers: $2^2 \times 3^2 = 36$.

Let's apply this to a few more examples. The GCF of 24 and 36 is 12, since 12 is the largest number that divides both. The LCM of 8 and 12 is 24, which is the smallest number both can divide into. For 14 and 28, we see that $14 = 2 \times 7$ and $28 = 2^2 \times 7$, so the GCF is 14 and the LCM is 28. Mastering GCF and LCM will help you with fractions, ratios, and solving many real-world problems.

1.7 Tip for Students / 유용한 팁

Math is not just about solving problems. It is about seeing how ideas fit together. Learn the basics—place value, fractions, comparisons, factors, primes, and multiples. They are not just chapters. They are tools. And the better you understand them, the more confidently you can use them anywhere.

Interesting Facts / 흥미로운 사실

- The word “digit” comes from the Latin word for finger, because people used their fingers to count.
- The decimal system is called “base-10” because each place is ten times the value of the place to its right.
- The largest single-digit number in the decimal system is 9.
- The introduction of zero by Indian mathematicians made calculations much clearer and easier.

1.8 Additional Concepts / 추가 개념

This section covers essential vocabulary and techniques that will help you master the fundamental concepts of numbers and their relationships. These additional tools complement the main concepts and provide practical methods for solving problems.

1.8.1 Place Value Vocabulary / 자리값 용어

Understanding place value requires familiarity with specific terminology. The essential place values you need to learn include billions, hundred millions, ten millions, millions, hundred thousands, ten thousands, thousands, hundreds, tens, and ones.

“한국어의 자리값 체계와 비슷하게, 왼쪽으로 갈수록 10배씩 커진다. 예: 일, 십, 백, 천, 만, 십만, 백만, 천만, 억, 십억, 백억”

“ones(일), tens(십), hundreds(백), thousands(천), ten thousands(만) 영어는 천부터 $\times 10$ ”

For decimal place values, the key terms are tenths, hundredths, thousandths, ten-thousandths, and millionths.

“소수의 자리값은 뒤에 ‘th’를 붙여 나타낸다. 예: tenths(소수 첫째 자리), hundredths(소수 둘째 자리), thousandths(소수 셋째 자리)”

“반드시 알아야하는 건 tenths(할), hundredths(푼), thousandths(리)”

1.8.2 Fraction Fundamentals

Fractions represent parts of a whole, and understanding their components is crucial. Every fraction has two main parts: the numerator and the denominator.

“분수(fraction)는 분자(numerator)와 분모(denominator)로 이루어져 있다.”

“외우기 힘들다면 AND라는 줄임말 기억하면 된다. AND는 분수 = 분자/분모(numerator/denominator)의 순서를 의미한다.”

There are different types of fractions that serve different purposes. An improper fraction has a numerator larger than its denominator, while a mixed number combines a whole number with a proper fraction.

“가분수(improper fraction)는 분자가 분모보다 큰 분수이다. 대분수(mixed number)는 자연수와 분수가 결합된 형태이다.”

“이는 성질만 이해하면 쉽게 외울 수 있다. 가분수는 분자가 분모보다 크므로 ‘부적절함(improper)’ 형태이고, 대분수는 자연수와 분수가 혼합된(mixed) 형태이다.”

1.8.3 Decimal Types / 소수의 종류

Decimals come in different forms, each with unique characteristics. A repeating or recurring decimal has a pattern that continues infinitely, while a finite decimal has a limited number of digits after the decimal point. An infinite decimal continues without end.

“순환소수(repeating/recurring)는 일정한 패턴이 끝없이 반복되는 소수이다. 유한소수(finite)는 소수 자리가 일정 부분에서 끝나는 소수이다. 무한소수(infinite)는 끝없이 이어지는 소수이다.”

“repeating/recurring(반복), in-(없다) + finite(제한이 있는) = infinite(무제한)”

“순환소수의 구별방법은 소수점위에 점의 유무 예: $1.\bar{3} = 1.333\dots$ 와 같이 위에 점으로 표시한다.”

“예: $1.\bar{3}47\bar{5}$ 에서 3과 5 위에 점이 찍혀 있다면, 그 구간이 순환되어 $1.347534753475\dots$ 이 된다.”

“무한소수는 분수로의 변환이 불가하다”

“순환소수는 분수로 변환할 수 있으며, 순환되는 자리수를 9로 나누어 표현한다. 예: $1.\overline{3984}$ 의 경우 $3984/9999 = 0.398439843984\dots$ 로 나타낼 수 있다.”

1.8.4 Comparison Vocabulary / 부등식 용어

Mathematics uses specific language to describe relationships between numbers. Understanding these terms helps you interpret problems correctly and communicate mathematical ideas clearly.

For greater than relationships, we use terms like greater than, more than, larger than, longer than, higher than, above, exceeds, and over. For greater than or equal to, we say greater than or equal to, at least, more than or equal to, minimum, no less than, not under/below, or not smaller than.

For less than relationships, we use less than, fewer than, smaller than, shorter than, lower than, under, below, or beneath. For less than or equal to, we say at most, does not exceed, maximum, no more than, up to, or not above/greater than.

“특히 at least(최소)와 at most(최대)는 반드시 알아두어야 한다. sat 수학에서 자주 사용된다.”

“무작정 외우기보다는 문제를 읽을 때 의미를 함께 생각해 보면 이해하기 쉽다.”

1.8.5 Prime Factorization / 소인수분해

Prime factorization is the process of breaking down a number into its smallest building blocks—prime factors. This technique is fundamental for understanding number structure and solving problems involving factors and multiples.

“prime factorization(소인수분해) 는 말 그대로 수를 소수(prime number)로 완전히 분해하는 것을 의미한다.”

“소인수분해의 기본 순서는 먼저 어떤 수를 분해할지 확인하는 것이다.”

“그 수가 어떤 배수인지 확인한 후 소수로 나누어 간다.”

To determine if a number is divisible by certain factors, we can use these rules:

- 2의 배수: 짝수/일의 자리 숫자가 2, 4, 6, 8, 0인 수
- 3의 배수: 각 자리의 수의 합이 3의 배수
- 끝의 두 자리 수가 00이거나 4의 배수
- 5의 배수: 일의 자리 숫자가 0또는 5
- 7의 배수: 세자리수 abc 일 때 $ab-c \times 2$ 가 7의 배수인수 (자주 사용되지는 않는다)

“위의 규칙으로 배수를 판단한 후, 작은 소수부터 차례대로 나누어 소인수분해를 진행한다.”

1.8.6 Finding GCF and LCM

The greatest common factor (GCF) and least common multiple (LCM) are essential concepts for working with multiple numbers. While these ideas may seem complex, there's a simple method using the Korean "ㄴ" technique that makes finding them straightforward.

“한국 학교에서는 ‘ㄴ’자 방법을 사용한다.”

“한국 학교에서 이미 배웠겠지만, ‘ㄴ’자 방법을 사용하면 쉽게 구할 수 있다.”

Let's find the GCF of 12 and 18 using this method:

$$\begin{array}{r|rr} 6 & 12 & 18 \\ & 2 & 3 \end{array}$$

“마지막에 남은 숫자가 모두 소수가 되면 소인수분해가 완료된다.”

“최대공약수(GCF)는 ‘ㄴ’자 옆에 있는 숫자들의 곱으로 구한다. 예: 6.”

Now let's find the LCM of 12 and 18:

$$\begin{array}{r|rr} 6 & 12 & 18 \\ & 2 & 3 \end{array}$$

“최소공배수(LCM)는 ‘ㄴ’자 옆과 밑에 있는 모든 숫자들의 곱으로 구한다. 예: $6 \times 2 \times 3 = 36$.”

These methods provide efficient ways to work with factors and multiples, making complex problems more manageable and helping you develop a deeper understanding of number relationships.

Chapter 2

Algebra and Variables / 대수와 변수

Welcome to the world of algebra! In this chapter, we'll learn how to use letters like **x** and **y** to represent numbers. These letters are called **variables**, and they help us write expressions and solve equations. Algebra might seem like a new language, but once you understand the rules, it can actually make math easier. Let's explore how we can describe patterns, solve problems, and think logically using simple algebraic tools.

Variables are like mystery boxes that can hold different numbers. When we write expressions like $3x + 5$, we're creating a rule that works for any value of x . This powerful idea helps us solve problems where we don't know the answer yet, but we know how things relate to each other.

2.1 Variables and Expressions / 변수와 식

Numbers are not always enough. Sometimes we don't know the exact value we are working with, but we still need a way to describe it. That's where variables come in. A variable is a symbol, usually a letter, that can stand for a number. With variables, we can write ideas as expressions instead of full sentences.

Think of $3x + 5$. It doesn't just mean "a number." It means something specific: three times a number, then add five. If $x = 2$, the expression becomes $3(2) + 5 = 11$.

Variables also let us describe real life. If you earn \$10 for every hour you work, you don't have to keep writing "10 times the number of hours." You can just write $10h$. When $h = 5$, that becomes 50. This is the power of algebra: turning words into symbols we can calculate

"A variable is a symbol, usually a letter, that can stand for a number."

변수(x, y, z 등)는 보통 문자로 표현되며, 어떤 수를 대신하여 나타낸다. 예: $x=8$ 일 때 $7x=56$.

"An expression is a mix of numbers, variables, and operations."

식(expression)은 숫자, 문자(변수), 그리고 연산(+, -, $\times$, $\div$)이 섞여 있는 수학적 표현이다

Examples:

$$3x + 5, \quad 7y - 2, \quad 10h$$

2.1.1 What are Variables and Expressions? / 변수와 식이란?

A **variable** is a placeholder for numbers. An **expression** shows a relationship using variables and numbers.

Examples:

- $3x + 5$
- $7y - 2$
- $10h$

Each expression can change depending on the value of the variable.

2.2 Simplifying Expressions / 식 약분하기

Expressions can look messy, but we can simplify them. Simplifying means combining like terms and applying the distributive property.

Like terms are terms that share the same variable. $2x + 3x$ is really $5x$. The distributive property is another tool: $3(x + 4)$ is the same as $3x + 12$. These two ideas—combining and distributing—make expressions shorter and easier to handle.

“Like terms are terms that share the same variable.”

동류항(like terms)은 같은 문자를 포함하는 항이다.

Example / 예시:

$2x, 3x$ 는 동류항이며, 합쳐서 $5x$ 가 된다.

“The distributive property is another tool: $3(x + 4)$ is the same as $3x + 12$.”

분배법칙(distributive property)은 괄호를 풀 때 사용된다.

Example / 예시:

$$3(x + 4) = 3x + 12$$

2.2.1 What does it mean to Simplify? / 약분이란?

To simplify an expression, you:

1. Combine like terms (same variable and power). 동류항을 정리한다.
2. Use distribution to remove parentheses. 분배법칙을 적용한다.

Example / 예시:

$$2x + 3x = 5x, \quad 4(y + 2) = 4y + 8$$

Simplifying makes expressions clear and prepares them for solving equations.

2.3 Solving One-Step Equations / 일차방정식 풀기

Equations are like balances. Whatever you do to one side, you must do to the other. One-step equations are the simplest kind. You undo the operation and find the value of the variable.

If $x + 4 = 9$, subtract 4 from both sides. $x = 5$. If $5x = 20$, divide both sides by 5. $x = 4$.

These may look small, but they show the heart of algebra: solving for the unknown.

“Equations are like balances. Whatever you do to one side, you must do to the other.”

방정식(equation))은 저울과 같다. 한쪽에 연산을 하면, 반대쪽도 똑같이 해야 균형이 유지된다.

2.3.1 What are One-Step Equations? / 일차방정식이란?

A one-step equation is solved with just one move—addition, subtraction, multiplication, or division.

“A one-step equation is solved with just one move—addition, subtraction, multiplication, or division.”

one-step equation(1차 방정식)은 한 번의 연산으로 해결되는 식이다.

Example / 예시:

$$x + 7 = 12 \rightarrow x = 5, 9y = 27 \rightarrow y = 3$$

Always check by plugging your answer back into the equation

2.4 Solving Multi-Step Equations / 다단계 방정식

Not all equations are simple. Some take more than one step. To solve them, we clean each side first, then isolate the variable step by step.

1. Take $2(x + 3) = 14$.
2. First, distribute: $2x + 6 = 14$.
3. Then subtract 6: $2x = 8$.
4. Finally, divide by 2: $x = 4$.

“Not all equations are simple. Some take more than one step.”

모든 방정식이 쉽게 풀리지는 않는다. 일차 이상의 방정식은 여러 단계를 거쳐야만 풀수 있다.
Step by step, the problem shrinks until the answer appears.

2.4.1 What are Multi-Step Equations? / 다단계 방정식이란?

A multi-step equation needs distribution, combining like terms, or more than one operation. You keep simplifying until the variable is alone.

“A multi-step equation needs distribution, combining like terms, or more than one operation.”

다단계 방정식(multi-step equation)은 분배법칙, 동류항 정리, 또는 여러 연산을 차례대로 적용하여 풀 수 있는 방정식이다.

Example:

1. $2(x + 3) = 14$
2. $2x + 6 = 14$
3. $2x = 8$
4. $x = 4$

This process teaches patience and logic.

2.5 Inequalities / 부등식

Equations tell us one exact value. Inequalities tell us ranges. Instead of saying $x = 5$, we might say $x > 5$. Inequalities are rules. They set boundaries.

The signs are:

- $>$: greater than
- $<$: less than
- $\geq$: greater than or equal to
- $\leq$: less than or equal to

“Inequalities tell us ranges. Instead of saying $x = 5$, we might say $x > 5$.”

inequality(부등식)은 값의 범위를 나타낸다. 예: $x = 5$ 대신 $x > 5$ 라고 하면, x 는 5보다 큰 모든 값을 가질 수 있다.

Solving inequalities is almost the same as solving equations—except for one key rule: if you multiply or divide by a negative number, you flip the sign.

“If you multiply or divide by a negative number, you flip the sign.”

부등식에서 음수로 곱하거나 나눌 때는 부등호의 방향을 바꾸어야 한다.

Examples:

- $x - 3 > 2 \Rightarrow x > 5$
- $-2x < 6 \Rightarrow \text{divide by } -2, \text{ flip} \Rightarrow x > -3$

2.5.1 What are Inequalities? / 부등식이란?

An inequality shows that values can be greater, smaller, or equal within a range.

For example, if a rule says “under 18,” we write $x < 18$. This describes everyone younger than 18.

Inequalities help us model rules, limits, and real-life conditions.

2.6 Tips for Students / 유용한 팁

Algebra is like learning a new language. At first the symbols look strange, but each step has meaning. Focus on variables, practice simplifying, and learn to solve equations patiently. These tools are not just for school—they're used everywhere in science, business, and everyday problem-solving.

2.6.1 Interesting Facts / 흥미로운 사실

- The word **algebra** comes from Arabic *al-jabr*, meaning “reunion of broken parts.” algebra(대수학)라는 단어는 아랍어 *al-jabr*에서 유래되었으며, “부서진 부분의 재결합”이라는 뜻이다.
- The balance scale is one of the oldest symbols of equations. Ancient mathematicians used it to explain equality. 저울은 방정식을 나타내는 가장 오래된 상징 중 하나로, 고대 수학자들은 이를 사용해 등식을 설명하였다.
- Inequalities appear in everything from sports scores to computer code. 부등식은 스포츠 점수부터 컴퓨터 코드까지 다양한 곳에서 나타난다.

2.7 Additional Concepts / 추가 개념

This section covers essential vocabulary and techniques that will help you master algebra. These tools complement the main concepts and provide strategies for solving equations and inequalities.

2.7.1 PEMDAS / 연산 순서

PEMDAS = Parentheses, Exponents, Multiplication/Division, Addition/Subtraction. 계산 순서를 외울 때 사용하는 영어 약어이다.

“미국에서는 괄호 → 지수 → 곱셈/나눗셈 → 덧셈/뺄셈 순서로 계산한다. 한국에서도 동일한 순서를 가르친다.”

“예: $2 + 3 \times 4 = 14$ (곱셈 먼저). 만약 괄호가 있다면 $(2 + 3) \times 4 = 20$.”

2.7.2 Transposition / 이항

Moving a term across the equals sign changes its sign.

Examples:

1. $3x+7=-3x-5$

2. $12=-6x$

3. $x=-2$

“이항할 때는 부호가 바뀐다. 곱셈과 나눗셈도 이항할 수 있다.”

Examples:

1. $\frac{3x}{7} = 6$

2. $3x = 42$

3. $x = 14$

2.7.3 Steps Order / 풀이 순서

한국식: 괄호 풀기 → 동류항 정리 → 이항 → 계수 나누기. 미국식: 식을 단순화(simplify) → 변수를 고립(isolate)한다.

“좌변(왼쪽)과 우변(오른쪽)이라는 용어는 한국 수학에서 자주 사용된다.”

Examples:

1. $2(x+3)+4=18$

2. $2x+6+4=18$

3. $2x+10=18$

4. $2x=8$

5. $x=4$

2.7.4 Graphing Inequalities / 부등식 그래프

When graphing inequalities on a number line:

- Open circle = not included ($<$, $>$)
- Closed circle = included ($\leq$, $\geq$)

“부등식을 수직선에 나타낼 때는 점과 화살표로 범위를 표시한다. 열린 원은 포함하지 않고, 닫힌 원은 포함한다.”

In two variables, inequalities shade one side of the boundary line.

Example:

- $y > 2x+1$ means shade above the line $y = 2x+1$.

“2차원에서는 직선을 경계로 하여 한쪽 영역을 음영 처리한다.”

2.7.5 Properties of Equality / 등식의 성질

Equations are like a balance. If you change one side, you must do the same to the other. 등식은 저울과 같아서, 한쪽을 바꾸면 다른 쪽도 똑같이 바뀌어야 한다.

- If $a = b$, then $a + c = b + c$. 양변에 같은 수를 더한다.
- If $a = b$, then $a - c = b - c$. 양변에 같은 수를 빼다.
- If $a = b$, then $ac = bc$. 양변에 같은 수를 곱한다.
- If $a = b$, then $a/c = b/c$. ($c \neq 0$) 양변을 같은 수로 나눈다.

2.7.6 Checking Solutions / 해 검증하기

After solving, plug the value back into the equation to check. 방정식의 해를 구한 후, 원래 식에 대입하여 맞는지 확인한다.

Example: $2x + 3 = 11 \Rightarrow x = 4$. Check: $2(4) + 3 = 11$ (True).

2.7.7 Common Mistakes / 자주 하는 실수

- $3x + 2y \neq 5xy$ (동류항이 아니므로 합칠 수 없다)
- Forgetting to flip inequality when dividing by negatives (음수로 나눌 때는 부등호를 반드시 바꿔야 한다)
- Dropping negative signs (계산 과정에서 부호를 잃어버리는 경우)

Chapter 3

Ratios, Rates, and Percents / (비, 비율, 백분율)

Welcome to Unit 3! In this chapter, you'll learn how to compare amounts using *ratios*, measure how fast things change with *rates*, and work with *percents* in real-life situations. These ideas are not just math—they show up in shopping, cooking, sports, and money. Ratios, rates, and percents help us describe relationships between numbers in clear and simple ways.

Ratios help us compare two quantities, like the number of boys to girls in a class or the ingredients in a recipe. Rates measure how things change over time or distance, such as speed in miles per hour or price per item. Percents give us a way to express parts of a whole, making it easy to understand discounts, grades, and statistics.

3.1 Ratios / 비

A ratio is about comparison. It tells us how much of one thing there is compared to another. Ratios are everywhere: the number of players on a team, the ingredients in a recipe, or the colors mixed for paint. A ratio doesn't just give numbers—it shows balance.

If you have 2 apples and 3 oranges, the ratio of apples to oranges is 2 to 3. You can write it $2 : 3$, $\frac{2}{3}$, or “2 to 3.” They all mean the same thing.

Ratios also show part-to-whole relationships. If a class has 12 boys and 8 girls, the ratio of boys to girls is $12 : 8$, which simplifies to $3 : 2$. But the ratio of girls to total students (20) is $8 : 20$, or $2 : 5$. Ratios let us see relationships clearly.

“A ratio is about comparison. It tells us how much of one thing there is compared to another.”

ratio(비)는 두 수나 양을 비교하는 방법이다. 예시: 사과 2개와 오렌지 3개의 비는 2:3

“Ratios also show part-to-whole relationships.”

비는 부분과 전체의 관계를 나타낼 수도 있다. 예시: 8명의 여학생 / 전체 20명 = 8:20

3.1.1 What is a Ratio? / 비란?

A **ratio** compares two numbers or quantities. It tells us how much of one thing there is compared to another. Ratios help us describe relationships and patterns, and they appear in recipes, sports, and even classrooms.

Examples:

- **Part-to-Part:** apples to oranges $\rightarrow 2 : 3$.

- **Part-to-Whole:** apples to total fruit $\rightarrow 2 : 5$.

Ways to write ratios:

- With a colon: **2 : 3**
- As a fraction: $\frac{2}{3}$
- In words: **2 to 3**

All three ways mean the same thing.

Part-to-Part and Part-to-Whole:

- **Part-to-Part:** Compares two individual parts (e.g., apples to oranges $\rightarrow 2 : 3$).
- **Part-to-Whole:** Compares one part to the total (e.g., apples to total fruits $\rightarrow 2 : 5$).

Example: If a class has 12 boys and 8 girls:

- The ratio of boys to girls is **12 : 8** $\rightarrow$ simplified to **3 : 2**.
- The ratio of girls to total students (20) is **8 : 20** $\rightarrow$ simplified to **2 : 5**.

Ratios simplify comparisons and help us see patterns.

Ratios and Proportions / 비와 비례식

Ratio / 비

서로 다른 두 수의 크기를 비교하는 것을 비라고 한다. 비를 나타낼 때는 기호 “:”를 써서 **● : ▲**로 나타낸다. **● : ▲**는 “**● 대 ▲**”라고 읽는다.

Ratios are written with the symbol “:”, such as **● : ▲**. Comparing the sizes of two different numbers is called a ratio. **● : ▲** is read as “**● to ▲**.”

A ratio compares two quantities, written “ $a : b$ ” and read “ a to b ”.

Proportion (as Rate) / 비율 기준량에 대한 비교하는 양의 크기를 비율이라고 한다. 비율은 분수와 소수로 나타낼 수 있다.

Proportion (as a rate) shows how large one quantity is relative to a reference; write as fraction or decimal.

$$1 : 2 \rightarrow \frac{1}{2} = 0.5 = 50\%$$

(Natural : Natural) / (자연수 : 자연수) 양쪽의 두 자연수를 최대공약수로 나누어 나타낸다.

$$3 : 6 = (3 \div 3) : (6 \div 3) = 1 : 2, \quad 12 : 20 = (12 \div 4) : (20 \div 4) = 3 : 5$$

(Fraction : Natural) / (분수 : 자연수) 양쪽에 분수의 분모를 곱한 다음에 자연수의 비로 고쳐서 계산한다.

$$\frac{1}{3} : 4 = \left(\frac{1}{3} \cdot 3 \right) : (4 \cdot 3) = 1 : 12$$

(Decimal : Natural) / (소수 : 자연수) 소수의 자릿수에 따라 10, 100, 1000, ... 을 곱한 다음에 자연수의 비로 고친다.

$$0.8 : 10 = (0.8 \cdot 10) : (10 \cdot 10) = 8 : 100 = 2 : 25$$

(Fraction : Fraction) / (분수 : 분수) 두 분모의 최소공배수를 곱한 다음 자연수의 비로 고쳐 계산한다.

$$\frac{3}{4} : \frac{9}{10} = 3 \cdot 10 : 9 \cdot 4 = 30 : 36 = 5 : 6$$

(Decimal : Decimal) / (소수 : 소수) 두 소수 중 자릿수가 큰 소수의 자릿수에 따라 10, 100, 1000, ... 을 곱한 다음 자연수의 비로 고친다.

$$0.6 : 1.25 = (0.6 \cdot 100) : (1.25 \cdot 100) = 60 : 125 = 12 : 25$$

$$0.58 : 4 = (0.58 \cdot 100) : (4 \cdot 100) = 58 : 400 = 29 : 200$$

Proportion (Equation) / 비례식 비율이 같은 두 비를 동시에 나타낸 식을 비례식이라고 한다. 예: $1 : 2 = 3 : 6$

An equation of two equal ratios is a proportion: $1 : 2 = 3 : 6$.

Property of Proportion / 비례식의 성질 비례식에서 외항의 곱 = 내항의 곱이다. 이 성질을 이용해 비례식을 풀 수 있다.

In $a : b = c : d$, $a \cdot d = b \cdot c$.

Ratio Division / 비례배분 전체를 주어진 비로 배분하는 것을 비례배분이라고 한다. 예: ●을 3 : 4로 나누면 각각 $\frac{3}{7}$, $\frac{4}{7}$ 씩 나누어 준다.

Divide a total in the ratio $m : n$ as $\frac{m}{m+n}$ and $\frac{n}{m+n}$ of the whole.

$$28 \text{을 } 3 : 4 \text{로 비례배분한다. } 28 \times \frac{3}{3+4} = 28 \times \frac{3}{7} = 12, \quad 28 \times \frac{4}{3+4} = 28 \times \frac{4}{7} = 16$$

$$81 \text{을 } 4 : 5 \text{로 비례배분한다. } 81 \times \frac{4}{4+5} = 81 \times \frac{4}{9} = 36, \quad 81 \times \frac{5}{4+5} = 81 \times \frac{5}{9} = 45$$

$$\text{Divide 28 in the ratio 3:4. } 28 \times \frac{3}{3+4} = 28 \times \frac{3}{7} = 12, \quad 28 \times \frac{4}{3+4} = 28 \times \frac{4}{7} = 16$$

$$\text{Divide 81 in the ratio 4:5. } 81 \times \frac{4}{4+5} = 81 \times \frac{4}{9} = 36, \quad 81 \times \frac{5}{4+5} = 81 \times \frac{5}{9} = 45$$

3.2 Rates

A rate is a special kind of ratio. It compares different units. We use rates to measure speed, prices, and even how fast we type. A car's speed is miles per hour. A store's price is dollars per item. A printer's speed is pages per minute.

If you drive 100 kilometers in 2 hours, the rate is

$$\frac{100}{2} = 50$$

kilometers per hour. Rates make comparisons fair because they put everything on the same scale.

“A rate is a special type of ratio that compares two different kinds of units.”

rate(율)은 특별한 종류의 비로, 서로 다른 단위를 비교함 예시: $100 \text{ km} \div 2 \text{ 시간} = 50 \text{ km/h}$, 즉 시속 50킬로미터이다

“Rates make comparisons fair because they put everything on the same scale.”

율(rate)은 모든 비교를 같은 기준으로 만들어 준다. 예: 가격을 비교할 때 달러/개 단위로 계산하면 어떤 상품이 더 저렴한지 알 수 있다.

3.2.1 What is a Rate? / 비율이란?

A **rate** is a special type of ratio that compares two different kinds of units. For example, it can compare miles to hours, dollars to items, or words to minutes.

Examples:

- 60 pages in 3 minutes $\rightarrow$ 20 pages per minute.
- \$200 earned in 8 hours $\rightarrow$ \$25 per hour.
- If you drive 100 kilometers in 2 hours, the rate is $\frac{100}{2} = 50 \text{ km/h}$.

Common rates:

- **Speed:** miles per hour (mph), kilometers per hour (km/h).
- **Prices:** dollars per item (\$/item).
- **Typing:** words per minute (wpm).

Everyday Examples:

- A printer prints 60 pages in 3 minutes $\rightarrow$ Rate: **20** pages per minute.
- A worker earns \$200 for 8 hours $\rightarrow$ Rate: **\$25** per hour.

Rates help us measure and compare in real life.

3.3 Unit Rate / 단위율

Sometimes we want to know the amount “per one.” That’s called a unit rate. If 3 notebooks cost \$6, then each notebook costs \$2. Unit rates are powerful because they help us compare fairly—per item, per mile, per hour.

“Sometimes we want to know the amount ‘per one.’ That’s called a unit rate.”

단위율(unit rate)은 두 번째 수가 1이 되도록 계산한 율이다.

“예: 노트 3권이 \$6일 때, 권당 가격은 \$2 (3:6 → 1:2).”

“Unit rates are powerful because they help us compare fairly—per item, per mile, per hour.”

단위율(unit rate)은 공정한 비교를 가능하게 한다.

“예: 사과 가격은 개당으로, 속도는 1시간당으로 비교한다.”

When grocery shopping, you might see two boxes of cereal: one \$4 for 8 ounces, another \$6 for 12 ounces. Unit rate shows which is the better deal.

3.3.1 What is a Unit Rate? / 단위율이란?

A **unit rate** is a rate where the second number is 1. It tells you how much of something there is *per one unit* of something else.

Examples:

- 120 miles in 4 hours → 30 miles per hour.
- \$4 for 8 apples → \$0.50 per apple.
- If 3 notebooks cost \$6, the unit rate is $\frac{6}{3} = \$2$ per notebook.

How to find a unit rate:

1. Divide the first number by the second.
2. Make sure the second number becomes 1.

Example:

$$120 \text{ miles in 4 hours} \rightarrow 120 \div 4 = 30 \text{ miles per hour.}$$

$$8 \text{ apples for } \$4 \rightarrow \$4 \div 8 = \$0.50 \text{ per apple.}$$

Real-life uses:

- Grocery stores list unit prices on shelf tags to help shoppers compare.
- Runners track their pace (minutes per mile) to measure speed.
- Restaurants calculate price per serving to set menu prices.

Unit rates appear in budgets, shopping, sports, and cooking.

3.4 Proportions / 비례식

Proportions show equality between two ratios. They tell us that two comparisons are balanced. If a recipe for 2 people uses 1 cup of rice, then a recipe for 4 people uses 2 cups. That's a proportion: the ratio stays the same.

We solve proportions by cross multiplying. If one ratio equals another, their cross products match. This makes it easy to find missing values.

“Proportions show equality between two ratios.”

비례식(proportion)은 두 비가 서로 같음을 나타낸다. 예: $\frac{2}{3} = \frac{4}{6}$.

“We solve proportions by cross multiplying.”

비례식은 교차 곱셈(cross multiplication)으로 풀 수 있다. 예: $\frac{x}{5} = \frac{6}{10} \Rightarrow 10x = 30 \Rightarrow x = 3$.

3.4.1 What is a Proportion? / 비례식이란?

A **proportion** is a statement that shows two ratios or rates are *equal*. It helps us solve problems where quantities grow or shrink at the same rate.

Example:

$$\frac{2}{3} = \frac{4}{6}$$

If $\frac{2}{3} = \frac{4}{6}$, this is a proportion because both sides are equal.

How to solve proportions: Use *cross multiplication*:

$$\frac{2}{3} = \frac{4}{6} \implies 2 \times 6 = 3 \times 4 \implies 12 = 12.$$

This means it is a true proportion.

You can also use proportions to find unknown numbers. For example,

$$\frac{x}{5} = \frac{6}{10} \implies 10x = 30 \implies x = 3.$$

Everyday examples:

- A map says 1 inch = 50 miles $\rightarrow$ use this ratio to calculate real distances.
- A recipe that serves 2 people can be doubled to serve 4 using proportions.

Proportions keep comparisons fair and balanced.

3.5 Percents / 백분율

A percent means “per hundred.” It is another way to compare numbers. 50% is half, 25% is a quarter, and 100% is the whole. Percents are part of everyday life—on price tags, test scores, and bank interest.

Converting between fractions, decimals, and percents is simple once you know the rules. Fractions turn into percents by multiplying by 100. Decimals turn into percents by moving the decimal two places to the right. Percents turn back into decimals by moving the decimal two places to the left.

“A percent means ‘per hundred.’”

percent(백분율)은 ‘100분의 ’를 의미한다. 예: $50\% = \frac{50}{100} = \frac{1}{2}$.

“Fractions turn into percentages by multiplying by 100.”

분수를 백분율로 바꾸려면 100을 곱하면 된다. 예: $\frac{1}{4} = 0.25 \Rightarrow 25\%$.

3.5.1 What is a Percent? / 백분율이란?

A **percent** shows part of a whole, out of 100. It is a number that means “*per 100*.” It shows a part out of a whole divided into 100 equal parts.

Examples:

- 50% means 50 out of 100 $\rightarrow \frac{1}{2}$.
- 25% means 25 out of 100 $\rightarrow \frac{1}{4}$.
- 100% means all of something.

3.5.2 Conversions

- Fraction $\rightarrow$ Percent: $\frac{1}{4} = 0.25 = 25\%$.
- Decimal $\rightarrow$ Percent: $0.6 = 60\%$.
- Percent $\rightarrow$ Decimal: $45\% = 0.45$.

How to convert:

- **Fraction to Percent:** multiply by 100.
 $\frac{1}{4} = 0.25 \Rightarrow 25\%$.
- **Decimal to Percent:** move the decimal point two places to the right.
 $0.6 \Rightarrow 60\%$.
- **Percent to Decimal:** move the decimal point two places to the left.
 $45\% \Rightarrow 0.45$.

3.5.3 Real-Life Examples

- **Shopping:** 30% off a jacket means you save 30 out of every 100 dollars.
- **Grades:** A score of 85% means you got 85 out of 100 points.
- **Interest Rates:** Banks use percents to calculate how much money you earn or owe.

Percents help us see proportions quickly and clearly.

3.6 Tip for / 유용한 팁

Ratios, rates, proportions, and percents are not just classroom topics. They describe how things relate and change. Learn to simplify ratios, find unit rates, solve proportions, and convert percents. These tools will help you shop smarter, plan better, and understand the world with clarity.

3.6.1 Interesting Facts / 흥미로운 사실

- Ratios are used in recipes, sports, and even mixing paint. 비는 요리, 스포츠, 색 혼합 등 다양한 곳에서 사용된다.
- Unit prices in stores are unit rates—they show cost per ounce or per item. 상점의 단가(unit price)는 단위율로, 온스당 또는 개당 가격을 나타낸다.
- The percent sign (%) was first used in Europe hundreds of years ago. Today, percents are global, from discounts to statistics. 백분율 기호(%)는 수백 년 전 유럽에서 처음 사용되었으며, 오늘날에는 할인부터 통계까지 전 세계적으로 쓰인다.

3.7 Additional Concepts / 추가 개념

This section introduces extra tools and ideas that connect with ratios, rates, and percents. These concepts provide shortcuts and strategies that make problem-solving easier.

3.7.1 Ratios and Fractions / 비와 분수

A ratio can be written as a fraction since both describe part-to-part or part-to-whole relationships.

“비는 부분과 전체의 관계를 나타내므로 분수로 변환할 수 있다. 예: $2 : 3 = \frac{2}{3}$.”

“비도 분수와 같이 약분할 수 있다. 예: $8 : 20 = 4 : 10 = 2 : 5$.”

“ $a : b = c : d$ 일 경우 $ad = bc$ 이다. 예: $4 : 13 = 8 : 26 \Rightarrow 4 \times 26 = 8 \times 13$.”

“이 원리를 활용하여 문제를 풀 수 있다. 예: $2 : 7 = 5 : x \Rightarrow 7 \times 5 = 35$, $2x = 35 \Rightarrow x = 17.5$.”

3.7.2 Distance, Speed, Time / 거리 · 속도 · 시간

The relationship between distance, speed, and time is often tested.

$$\text{Distance} = \text{Speed} \times \text{Time}, \quad \text{Speed} = \frac{\text{Distance}}{\text{Time}}, \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

“거리는 속도 $\times$ 시간, 속도는 거리/시간, 시간은 거리/속도이다.”

3.7.3 Metric Prefixes / 미터법 접두사

Common prefixes in the metric system are useful for converting units.

$10^9 = \text{G}$ (giga), $10^6 = \text{M}$ (mega), $10^3 = \text{k}$ (kilo), $10^{-2} = \text{c}$ (centi), $10^{-3} = \text{m}$ (milli), $10^{-6} = \mu$ (micro), $10^{-9} = \text{n}$ (nano)

“예: $1\text{m} = 1000\text{mm}$, $1\text{mm} = 0.001\text{m}$.”

3.7.4 Complex Fractions / 번분수

Complex fractions appear often when working with unit rates. To simplify:

$$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

“번분수는 분자와 분모를 교차 곱하여 계산한다.”

Example: \$19/4 for 5 books $\rightarrow (19/4) \div (5/1) = 19/20$ per book.

3.7.5 Butterfly Method / 버터플라이 메서드

Another name for cross multiplication.

$$\frac{x}{5} = \frac{8}{2} \Rightarrow 2x = 40 \Rightarrow x = 20$$

“교차 곱셈의 모양이 나비와 비슷하여 butterfly method라고 부른다.”

3.7.6 Decimals and Percentages / 소수와 백분율

Decimals and percentages are interchangeable:

$$\text{Decimal} \times 100 = \text{Percentage}, \quad \text{Percentage} \div 100 = \text{Decimal}$$

“소수에 100을 곱하면 백분율이 되고, 백분율을 100으로 나누면 소수가 된다.”

Example problem: A shirt costs \$100. It's increased by 30% and then discounted by 50%.

$$100 \times (1 + 0.3) \times (1 - 0.5) = 100 \times 1.3 \times 0.5 = 65$$

“최종 가격은 \$65이다.”

Appendix / 부록

Important Formulas and Definitions / 중요한 공식과 정의

- **Place Value / 자리값:** Value of a digit = digit $\times$ place value. 숫자의 위치에 따라 값이 정해진다. 예: $4275 = 4000 + 200 + 70 + 5$
- **Fraction to Decimal / 분수를 소수로:** $\frac{a}{b} = a \div b$. 분수는 나눗셈으로 변환할 수 있다.
- **Prime Number / 소수:** A number greater than 1 with exactly two factors: 1 and itself. 1과 자기 자신만 약수로 갖는 수이다.
- **Composite Number / 합성수:** A number with more than two factors. 약수가 두 개보다 많은 수이다.
- **GCF / 최대공약수:** Product of the lowest powers of common primes. 공통 소인수의 가장 낮은 지수를 곱한 값이다.
- **LCM / 최소공배수:** Product of the highest powers of all primes present. 모든 소인수의 가장 높은 지수를 곱한 값이다.

Further Reading and Resources / 추가 학습 자료

- "Elementary Mathematics" by Liping Ma
- "How to Be a Math Genius" by Mike Goldsmith
- Khan Academy: <https://www.khanacademy.org/>
- Purplemath: <https://www.purplemath.com/>

Summary of Key Concepts / 핵심 개념 요약

Chapter 1: Numbers & Operations / 1단원: 수와 연산

- **Place Value / 자리값**: Each digit's value depends on its position. Example: $4275 = 4000 + 200 + 70 + 5$.
- **Decimals and Fractions / 소수와 분수**: Both represent parts of a whole. Example: $0.75 = \frac{3}{4}$.
- **Comparing and Ordering / 수의 비교와 정렬**: Use $>$, $<$, $=$; convert to decimals for easier comparison. Example: $\frac{2}{5} < \frac{3}{7}$.
- **Factors and Multiples / 약수와 배수**: Factors divide evenly; multiples are products. Example: factors of 12 are 1, 2, 3, 4, 6, 12.
- **Prime vs. Composite / 소수와 합성수**: Primes have exactly two factors (1 and itself); composites have more. Example: 2, 3, 5 are prime; 4, 6, 8 are composite.
- **GCF and LCM / 최대공약수와 최소공배수**: GCF = largest common factor; LCM = smallest common multiple. Example: for 12, 18, GCF = 6, LCM = 36.

Chapter 2: Algebra / 2단원: 대수

- **Variables & Expressions / 변수와 식**: Variables represent numbers; expressions combine numbers, variables, and operations.
- **Simplifying Expressions / 식 단순화**: Combine like terms and use the distributive property.
- **One-Step Equations / 일차(한 단계) 방정식**: Solved with one operation; apply the same to both sides.
- **Multi-Step Equations / 다단계 방정식**: Use distribution $\rightarrow$ combine like terms $\rightarrow$ isolate the variable.
- **Inequalities / 부등식**: Show ranges; flip the inequality sign when multiplying/dividing by a negative.

Chapter 3: Ratios, Rates, and Percents / 3단원: 비, 율, 백분율

- **Ratios / 비**: Compare two numbers or quantities (part-part, part-whole).
- **Rates / 율**: Ratios with different units (e.g., speed, cost per item).
- **Unit Rate / 단위율**: Rate with denominator 1 (e.g., per item, per hour).
- **Proportions / 비례식**: Equality of two ratios; solved by cross multiplication.
- **Percents / 백분율**: Mean "per hundred"; easily converted with fractions/decimals.

Final Words / 맺음말

Congratulations on completing this book! You've worked through numbers, algebra, ratios, percents, and more. That's not easy—especially while learning in two languages.

“이 책을 끝까지 마친 여러분을 축하합니다! 수, 대수학, 비와 백분을 등 다양한 개념을 공부했습니다. 두 언어로 동시에 배우는 일은 쉽지 않지만, 여러분은 해냈습니다.”

Remember, math is like a language. The more you practice, the more fluent you become. Mistakes are part of learning, so don't be discouraged.

“수학은 하나의 언어와 같습니다. 연습할수록 더 유창해집니다. 실수는 배움의 과정이니 절대 낙심하지 마세요.”

Next Steps / 다음 단계

- Keep practicing problems every day, even small ones.
- Review the vocabulary lists in this book regularly.
- Explore extra resources like Khan Academy or Purplemath.
- Challenge yourself with harder questions step by step.

“매일 조금씩 문제를 풀고, 책 속 어휘를 반복 복습하며, 온라인 자료도 활용하세요. 그리고 한 단계씩 더 어려운 문제에 도전해 보세요.”

A Personal Note / 개인적인 메시지

We wrote this book because we once struggled too. If you ever feel stuck, remember—you are not alone. Step by step, you'll grow stronger in both math and English. One day, you'll be the one helping others.

“저희도 같은 어려움을 겪었기에 이 책을 만들었습니다. 힘들 때가 있어도 혼자가 아님을 기억하세요. 차근차근 연습하다 보면 수학과 영어 모두에서 성장할 것입니다. 언젠가는 여러분이 다른 사람을 도와줄 차례가 올 것입니다.”

Thank you for learning with us. Keep going—you've got this! 끝까지 함께해 주셔서 감사합니다. 계속 나아가세요—여러분은 할 수 있습니다!

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- [2] *Help your kids with math: a unique step-by-step visual guide*. First American edition. New York, New York: DK Publishing, 2010. ISBN: 9780756649791.